Lecture 26

Equivariant line bundles and Borel–Weil

Objectives. *Lecture 12 induced from a character of a torus and obtained infinitely many irreducible representations. In this lecture a complex structure selects one of them. Induced representations act on equivariant functions, and the same functions can be read as sections of a vector bundle. We compute the holomorphic sections completely for the projective line. We then state the general Borel–Weil theorem, work the case* SU(3) *by hand, and prove the theorem from Peter–Weyl.*

By the end of this lecture you will be able to:

- Compute all holomorphic sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$.
- Identify the resulting SU(2) representation and its central action.
- State Borel–Weil with a specified bundle character and dual convention, and write explicit sections for SU(3).
- Prove Borel–Weil from Peter–Weyl, and name the inputs that stay cited: $G = KB$, the complex structure of G/B, the character χ_λ and the highest-weight theorem.

26.1 From equivariant functions to sections

Let G be a Lie group, H a closed subgroup and $\sigma : H \to \mathrm{GL}(V)$ a continuous finite-dimensional complex representation. The quotient projection $G \to G/H$ is a submersion by Lecture 3. So it has a smooth local section near every point of G/H, by the local-section theorem (Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.26). We now attach a copy of V above each coset.

Definition 26.1. The *associated vector bundle* is

$$E = G \times_H V = (G \times V)/\!\sim, \qquad (gh, v) \sim (g, \sigma(h)v).$$

Its projection is $[g, v] \mapsto gH$. A *section* is a map $s : G/H \to E$ whose value at x lies in the fibre above x. If $\dim V = 1$, it is a *line bundle*.

Each fibre is a vector space: choose g over the coset and use $v \mapsto [g, v]$. Changing g changes this identification by an invertible linear map.

Choose smooth local sections $s_i : U_i \to G$ whose domains cover G/H. The map

$$U_i \times V \longrightarrow E|_{U_i}, \qquad (x, v) \longmapsto [s_i(x), v]$$

is a bijection, because every g over x has the form $s_i(x)h$ with $h \in H$. It is linear on each fibre. On $U_i \cap U_j$, write $s_j(x) = s_i(x)h_{ij}(x)$. The same bundle vector then has coordinates

$$v_i = \sigma(h_{ij})v_j.$$

The identity $h_{ij}h_{jk} = h_{ik}$ ensures consistency on triple overlaps.

A *smooth complex vector bundle* of rank k over a manifold M is a smooth manifold E with a smooth surjection onto M and a complex vector-space structure on each fibre E_x, the preimage of x. It is required to be locally a product. Each point of M has an open neighbourhood U and a diffeomorphism $E|_U \to U \times \mathbb{C}^k$ mapping each fibre E_x complex-linearly onto $\{x\} \times \mathbb{C}^k$. These diffeomorphisms are the *smooth local trivialisations*.

These transition functions make $E = G \times_H V$ a smooth complex vector bundle, and this structure does not depend on the chosen family of local sections. The tool is the vector bundle chart lemma (Lee, Lemma 10.6). The section on sources at the end of this lecture checks its hypotheses.

A section is equivalently a function $f : G \to V$ with

$$f(gh) = \sigma(h)^{-1} f(g). \tag{26.1}$$

The correspondence is $s(gH) = [g, f(g)]$. It is well-defined because

$$[gh, f(gh)] = [g, \sigma(h)f(gh)] = [g, f(g)].$$

Conversely, a section determines a unique $f(g)$ in the coordinates supplied by g. Over U_i, the section is $x \mapsto [s_i(x), f(s_i(x))]$. The local trivialisation is a diffeomorphism, so s is smooth over U_i exactly when the coefficient $f \circ s_i$ is smooth. Moreover, s is smooth exactly when f is smooth, and s is continuous exactly when f is continuous. The section on sources gives the proof.

Left multiplication defines $a[g, v] = [ag, v]$. On sections its action is

$$(\Pi(a)s)(x) = a\, s(a^{-1}x).$$

In the function description this becomes $(\Pi(a)f)(g) = f(a^{-1}g)$. This is precisely the compact induction convention of Lecture 12.

26.2 The projective line and its two charts

We first compute everything on one example, and we start with the holomorphic language it needs. A function on an open subset of $\mathbb{C}^d$ is *holomorphic* if it is real C^∞ and its differential is $\mathbb{C}$-linear at every point. Polynomials are holomorphic, and so are quotients of polynomials where the denominator is nonzero. Composition of holomorphic maps is holomorphic by the chain rule.

A *complex manifold* is a Hausdorff, second-countable topological space covered by charts. These charts are homeomorphisms of open sets onto open subsets of $\mathbb{C}^d$. The transition maps between them have holomorphic components. A map between complex manifolds is *holomorphic* if it is continuous and its expressions in charts have holomorphic components.

Definition 26.2. The *complex projective line* $\mathbb{CP}^1$ is the set of one-dimensional complex subspaces of $\mathbb{C}^2$. A nonzero vector (v_0, v_1) determines the point $[v_0 : v_1]$, with $[cv_0 : cv_1] = [v_0 : v_1]$ for $c \neq 0$.

There are two charts. On $U_0 = \{v_0 \neq 0\}$ put $z = v_1/v_0$. On $U_1 = \{v_1 \neq 0\}$ put $w = v_0/v_1$. On the overlap, the transition between the charts is $w = 1/z$.

The charts give $\mathbb{CP}^1$ its topology. The points $[1 : 0]$ and $[0 : 1]$ have disjoint neighbourhoods $\{|z| < 1\}$ and $\{|w| < 1\}$. Every other pair lies in a common chart, so the space is Hausdorff. The two charts give a countable base. Each chart is a copy of $\mathbb{C}$, and the transition $w = 1/z$ is holomorphic on the overlap. Hence the two charts make $\mathbb{CP}^1$ a complex manifold, in the sense above.

The group $\mathrm{SU}(2)$ acts on lines by matrix multiplication. In the z chart, the action of $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ is

$$z \longmapsto \frac{b + \bar{a}z}{a - \bar{b}z}$$

where the denominator is nonzero. The other chart expressions are similar. These rational expressions show that the action is smooth. It acts transitively. Indeed, a unit vector $v = (a, b)$ is the first column of

$$g_v = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2).$$

The stabiliser of the line $\mathbb{C}(1, 0)$ is

$$T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})\}.$$

By the homogeneous-space theorem of Lecture 3, $\mathbb{CP}^1 \cong \mathrm{SU}(2)/T$ as smooth manifolds. It is compact, as the image of the compact group.

The *tautological line bundle* has fibre the line itself:

$$\mathcal{O}(-1) = \{(\ell, v) : \ell \in \mathbb{CP}^1,\ v \in \ell\}.$$

Its dual is $\mathcal{O}(1)$. For $m \geq 0$, let $\mathcal{O}(m) = \mathcal{O}(1)^{\otimes m}$. Set $\mathcal{O}(0)$ equal to the trivial line bundle. For $m < 0$, use the $(-m)$th tensor power of $\mathcal{O}(-1)$.

The local frames of $\mathcal{O}(-1)$ are $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. On the overlap, $e_1 = z^{-1}e_0$. The dual frames satisfy $e_1^* = ze_0^*$. For $m < 0$, $(e_i^*)^m$ means $e_i^{\otimes(-m)}$. Since $e_1^{\otimes(-m)} = z^m e_0^{\otimes(-m)}$, the same transition rule holds for $m < 0$. Hence a section of $\mathcal{O}(m)$ is a pair of functions f_0, f_1 with

$$f_0(z) = z^m f_1(1/z), \qquad f_1(w) = w^m f_0(1/w). \tag{26.2}$$

The frames $(e_0^*)^m$ and $(e_1^*)^m$ give fibrewise-linear bijections $\mathcal{O}(m)|_{U_i} \to U_i \times \mathbb{C}$. Their transition is multiplication by z^m, which is smooth and holomorphic on $U_0 \cap U_1$. The vector bundle chart lemma of Lee, Lemma 10.6, therefore makes $\mathcal{O}(m)$ a smooth line bundle with these trivialisations. We therefore call the section *holomorphic* when both functions are *entire*, that is, holomorphic on all of $\mathbb{C}$. The Borel–Weil discussion below shows that this definition agrees with the notion for associated bundles. We denote this space by $H^0(\mathbb{CP}^1, \mathcal{O}(m))$. The superscript 0 means ordinary global sections.

Since $w = 1/z$, the plane of z with one point added at infinity is all of $\mathbb{CP}^1$, so $\mathbb{CP}^1$ is a sphere. Write $E_0 = (e_0^*)^m$ and $E_1 = (e_1^*)^m$ for the two frames of $\mathcal{O}(m)$. They satisfy $E_1 = z^m E_0$ on the overlap. A section value over a point has one coefficient in each frame. If the frame grows by z^m, then the coefficient shrinks by z^{-m}. Figure 26.1 shows this at one point.

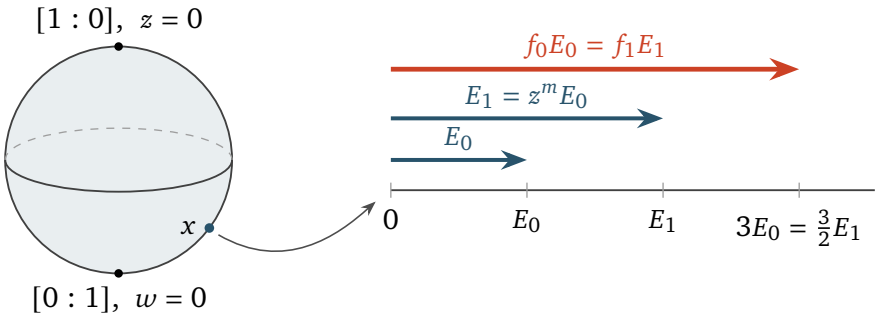


Figure 26.1: The sphere $\mathbb{CP}^1$ with the point x where $z = 2$ and $w = \frac{1}{2}$, and the fibre of $O(1)$ over x, drawn as a real line. On this sphere the chart z omits the south pole $[0 : 1]$, and the chart w omits the north pole $[1 : 0]$. Here $E_1 = 2E_0$, and the section value $3E_0$ has coefficient $f_0 = 3$ in the frame E_0 and coefficient $f_1 = \frac{3}{2}$ in the frame E_1.

26.3 Computing the holomorphic sections

We import two results from complex analysis, the power-series theorem and Cauchy's estimate. If f is entire, then

$$f(z) = \sum_{k \geq 0} a_k z^k, \qquad |a_k| \leq R^{-k} \max_{|z|=R} |f(z)| \quad (R > 0).$$

Both are imported from Stein and Shakarchi, *Complex Analysis* (2003), Chapter 2, Section 4. The section on sources records the exact statements. These results apply to our entire functions, because a $\mathbb{C}$-linear differential makes a function complex differentiable.

Theorem 26.3. *For $m \geq 0$, the holomorphic sections of $O(m)$ are naturally the homogeneous polynomials of degree m on $\mathbb{C}^2$. In particular,*

$$\dim H^0(\mathbb{CP}^1, O(m)) = m + 1.$$

For $m < 0$, the space of holomorphic sections is zero.

Proof. Let (f_0, f_1) satisfy the transition rule (26.2). Since f_1 is bounded on $|w| \leq 1$, there is C with

$$|f_0(z)| \leq C|z|^m \qquad (|z| \geq 1).$$

If $f_0(z) = \sum a_k z^k$, Cauchy's estimate on a circle of radius $R \geq 1$ gives $|a_k| \leq CR^{m-k}$. Let $R \to \infty$. For $k > m$ this forces $a_k = 0$.

If $m \geq 0$, it follows that f_0 is a polynomial of degree at most m. Conversely every such polynomial works, because

$$f_0(z) = \sum_{k=0}^{m} a_k z^k \quad \Longrightarrow \quad f_1(w) = \sum_{k=0}^{m} a_k w^{m-k}$$

is entire and satisfies the transition. The corresponding homogeneous polynomial is

$$F(v_0, v_1) = \sum_{k=0}^{m} a_k v_0^{m-k} v_1^k.$$

Its chart coefficients are $F(1,z) = f_0(z)$ and $F(w,1) = f_1(w)$. It obeys $F(cv) = c^m F(v)$ for $c \in \mathbb{C}$, so its restriction to each line is an element of that line's mth dual tensor power. Hence F is the coordinate-free description of the section.

If $m < 0$, every $k \geq 0$ satisfies $k > m$, so all a_k vanish. The transition then makes $f_1 = 0$ too. ∎

For $m = 2$, take $f_0(z) = a_0 + a_1 z + a_2 z^2$. Then $f_1(w) = a_0 w^2 + a_1 w + a_2$. Thus $F = a_0 v_0^2 + a_1 v_0 v_1 + a_2 v_1^2$.

26.4 The group action and its dual

A matrix $g \in \mathrm{SL}(2,\mathbb{C})$ sends a tautological fibre ℓ linearly to $g\ell$. The inverse transpose of this linear map sends the dual line ℓ^* to $(g\ell)^*$. Hence g acts on $O(1)$ and on all its tensor powers. Under the polynomial description, the section action is

$$(\Pi_m(g)F)(v) = F(g^{-1}v). \tag{26.3}$$

Indeed, at the line ℓ, the transformed functional evaluates a vector $v \in \ell$ by first pulling it back to $g^{-1}v \in g^{-1}\ell$. This gives the inverse in the formula.

The representation is $\mathrm{Sym}^m(\mathbb{C}^2)^*$, the dual of the symmetric tensor representation. For $\mathrm{SU}(2)$ this is isomorphic to V_m, by the prerequisite highest-weight theory. Here is also a direct weight check. On the basis $F_k = v_0^{m-k} v_1^k$,

$$\Pi_m(t_\theta)F_k = e^{i(2k-m)\theta}F_k, \qquad 0 \leq k \leq m.$$

The weights are $-m, -m+2, \ldots, m$. They match V_m, with highest weight m and dimension $m+1$. Irreducibility is the symmetric-power result from Lie Algebras.

For a concrete matrix computation, put

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

For $m = 2$, in the ordered basis $(v_0^2, v_0 v_1, v_1^2)$, expansion gives

$$D^{(2)}(g) = \begin{pmatrix} \bar{a}^2 & -\bar{a}b & b^2 \\ 2\bar{a}\bar{b} & |a|^2 - |b|^2 & -2ab \\ \bar{b}^2 & a\bar{b} & a^2 \end{pmatrix}.$$

For example, the middle column comes from

$$(\bar{a}v_0 + \bar{b}v_1)(-bv_0 + av_1) = -\bar{a}b\, v_0^2 + (|a|^2 - |b|^2)v_0 v_1 + a\bar{b}\, v_1^2.$$

The centre acts by $\Pi_m(-I) = (-1)^m I$. Thus the action on sections descends to $\mathrm{SO}(3)$ precisely for even m.

The same sections inside compact induction

For a homogeneous polynomial F of degree $m \geq 0$, a corresponding equivariant function $f_F : \mathrm{SU}(2) \to \mathbb{C}$ is given by $f_F(g) = F(g(1,0)^T)$. In the proof of Theorem 26.3, the section defined by F assigns to each line the restriction of F to that line. So $f_F(g)$ is the value of this restriction at the spanning vector $g(1,0)^T$. Since $gt_\theta(1,0)^T = e^{i\theta}g(1,0)^T$, homogeneity gives

$$f_F(gt_\theta) = e^{im\theta} f_F(g).$$

This is the covariance (26.1) for $H = T$ and $\sigma(t_\theta) = e^{-im\theta}$. Also

$$f_{\Pi_m(a)F}(g) = F(a^{-1}g(1,0)^T) = f_F(a^{-1}g),$$

so $F \mapsto f_F$ intertwines Π_m with the left-translation action $(\Pi(a)f)(g) = f(a^{-1}g)$ of the first section.

Next we identify the whole bundle. Let m be any integer. At the line $\mathbb{C}(1,0)$, the element t_θ acts on the tautological fibre by $e^{i\theta}$. It acts on the mth dual power of that fibre by $\chi_{-m}(t_\theta) = e^{-im\theta}$. Write $\mathbb{C}_{-m}$ for $\mathbb{C}$ with T acting through the character χ_{-m}.

Let ε be the value at $z = 0$ of the frame $(e_0^*)^m$. For $m < 0$, this frame means $e_0^{\otimes(-m)}$, consistently with $e_1^* = ze_0^*$ and $e_1 = z^{-1}e_0$. The map $[g,u] \mapsto g \cdot (u\varepsilon)$ from $\mathrm{SU}(2) \times_T \mathbb{C}_{-m}$ to $\mathcal{O}(m)$ is well defined, because t_θ multiplies ε by $\chi_{-m}(t_\theta)$. It is $\mathrm{SU}(2)$-equivariant and linear on each fibre. Since $\mathrm{SU}(2)$ acts transitively on $\mathbb{CP}^1$ with stabiliser T, it is bijective. We now show that its coefficients in the local trivialisations are smooth and nowhere zero.

Over U_0, use the smooth local section $s(z) = g_v$ of $\mathrm{SU}(2) \to \mathrm{SU}(2)/T$, where $v = (1,z)/\sqrt{1+|z|^2}$. Since $g_v(1,0)^T = v$, we have

$$s(z)^{-1}(1,z)^T = \sqrt{1+|z|^2}\,(1,0)^T.$$

The element $s(z)$ sends a functional ϕ on $\mathbb{C}(1,0)$ to $\phi \circ s(z)^{-1}$ on the line through $(1,z)$. Evaluating at $e_0(z) = (1,z)$ gives $s(z) \cdot e_0^*(0) = \sqrt{1+|z|^2}\, e_0^*(z)$. For vectors, the same display gives $s(z)e_0(0) = (1+|z|^2)^{-1/2}e_0(z)$. Taking mth or $(-m)$th tensor powers, we obtain for every integer m

$$s(z) \cdot (u\varepsilon) = u\,(1+|z|^2)^{m/2}\,(e_0^*)^m(z).$$

Over U_1, the section g_v with $v = (w,1)/\sqrt{1+|w|^2}$ gives the coefficient $u(1+|w|^2)^{m/2}$ with respect to $(e_1^*)^m$, by the same computation.

Thus, in both local trivialisations, the map is $u \mapsto u(1+|x|^2)^{m/2}$, where x is the chart coordinate. This coefficient is smooth. Its factor $(1+|x|^2)^{m/2}$ is nowhere zero, so the inverse $c \mapsto c(1+|x|^2)^{-m/2}$ is smooth too. Hence the map is an isomorphism of smooth line bundles

$$\mathcal{O}(m) \cong \mathrm{SU}(2) \times_T \mathbb{C}_{-m}.$$

A section in the equivariant-function description therefore obeys $f(gt_\theta) = e^{im\theta} f(g)$. The positive exponent belongs to the function covariance. The fibre itself has the negative character. For $m \geq 0$ and a homogeneous polynomial F of degree m, the vector $g \cdot (u\varepsilon)$ takes the value u at $g(1,0)^T$, by the definition of ε. With $u = f_F(g) = F(g(1,0)^T)$, it agrees with F at that vector, so the two descriptions of the section agree.

Now we can say what holomorphy does. The Hilbert completion of the continuous induced representation of Lecture 12 is the space of all L^2 sections of $\mathcal{O}(m)$. Let $m \geq 0$. By Frobenius

reciprocity, V_d occurs in this space once when $e^{-im\theta}$ is a weight of V_d, that is, for $d = m, m+2, m+4, \ldots$. Figure 26.2 shows this list. By Theorem 26.3, the holomorphic sections form exactly the first term V_m. For $m < 0$ the same reciprocity gives $V_{|m|}, V_{|m|+2}, \ldots$ while holomorphy keeps nothing.

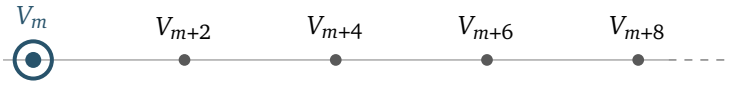


Figure 26.2: The irreducible representations of SU(2) in the L^2 sections of $O(m)$, for $m \geq 0$. Each dot occurs once. The ringed dot V_m is the space of holomorphic sections.

In physics language, $O(m)$ is the bundle of a charged particle on a sphere around a magnetic monopole, its two charts are two gauge patches, and the holomorphic sections are the lowest Landau level.

26.5 The general Borel–Weil theorem

We now replace $\mathrm{SL}(2,\mathbb{C})$ and its upper triangular subgroup by a complex group G and a Borel subgroup B. The SU(2) case at the end of this section shows that the calculation above is the first instance.

A *complex submanifold* of a complex manifold M is a subset N with the following property. Each point of N lies in a chart of M in which N is given by the vanishing of some coordinates. The remaining coordinates give charts of N. Hence a holomorphic map into M with values in N is also holomorphic as a map into N. We call this the *slice-chart property*.

A *complex Lie group* is a group with a complex manifold structure for which multiplication and inversion are holomorphic. The groups $K_{\mathbb{C}}$ of Lecture 14 are examples. A *closed complex subgroup* of a complex Lie group G is a closed subgroup that is also a complex submanifold.

Holomorphic bundles. Suppose that G is a complex Lie group, H is a closed complex subgroup and G/H is a complex manifold. Suppose also that the projection $\pi : G \to G/H$ is holomorphic, that σ is holomorphic and that π has holomorphic local sections s_i whose domains U_i cover G/H. Then $h_{ij} = s_i^{-1} s_j$ is holomorphic as a map into G, because the group operations of G are holomorphic. By the slice-chart property, h_{ij} is holomorphic as a map into H. Then $\sigma(h_{ij})$ is holomorphic, as a composition of holomorphic maps. So the local trivialisations given by the sections s_i make E a complex manifold. In this case E is a *holomorphic* vector bundle. These trivialisations are its *holomorphic bundle charts*, and the complex structure of E does not depend on the family (s_i); the section on sources checks these points. A section is *holomorphic* when each coefficient $f \circ s_i$ is holomorphic.

A section is holomorphic exactly when its function $f : G \to V$ is holomorphic. Indeed, if f is holomorphic, then so is $f \circ s_i$. Conversely, the map $q(g) = s_i(\pi(g))^{-1} g$ on $\pi^{-1}(U_i)$ is holomorphic into H by the slice-chart property. The covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big),$$

and both factors are holomorphic. Hence f is holomorphic on each set $\pi^{-1}(U_i)$, and therefore on G.

The groups. Let K be compact, connected, simply connected and semisimple. Its complexification $G = K_{\mathbb{C}}$ was introduced in Lecture 14, and it is connected because K is. Fix a maximal torus T of K and positive roots. The *positive Borel subalgebra* $\mathfrak{b}$ is the complex Cartan subalgebra plus the positive root spaces. Let B be the connected Lie subgroup of G with this Lie algebra.

We cite two facts about these groups. First, B is closed, and the inclusion of K in G induces a diffeomorphism from K/T onto the *flag manifold* G/B (Sepanski, *Compact Lie Groups*, Lemma 7.49 and Theorem 7.50). In particular,

$$G = KB.$$

Second, B is a closed complex subgroup, and G/B is a complex manifold with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections (Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 2.11 and its proof). So the paragraph on holomorphic bundles applies to G, B and G/B. The section on sources identifies Sepanski's matrix model of G with the group of Lecture 14, and it proves that B is a complex subgroup.

The character. Let λ be an integral weight. Since K is simply connected, $X^*(T) = P$ (Lecture 7). Let N_+ be the connected subgroup of G whose Lie algebra is the sum of the positive root spaces. Each $b \in B$ is uniquely a product $t\exp(iH)n$ with $t \in T$, $H \in \mathfrak{t}$ and $n \in N_+$, and these factors depend smoothly on b (Sepanski, Lemma 7.49(d)). Our character of weight λ on T is $\exp H \mapsto e^{i\lambda(H)}$. Define $\chi_\lambda : B \to \mathbb{C}^\times$ by

$$\chi_\lambda\big(\exp(H')\exp(iH)n\big) = e^{i\lambda(H')-\lambda(H)} \qquad (H, H' \in \mathfrak{t},\ n \in N_+).$$

We import from Sepanski, Definition 7.52, the fact that this formula defines a homomorphism. The formula is well defined on $T = \exp\mathfrak{t}$ because $\lambda \in P = X^*(T)$. By the smooth dependence of the factors on b, the homomorphism χ_λ is smooth. It is trivial on N_+. Its differential at e is the complex-linear extension of $i\lambda$ on $\mathfrak{t}_{\mathbb{C}}$ and zero on the positive root spaces, so it is $\mathbb{C}$-linear. Left translations of B are holomorphic, and χ_λ is multiplicative. So its differential is $\mathbb{C}$-linear at every point, and χ_λ is holomorphic.

The bundle. Define

$$L_\lambda = G \times_B \mathbb{C}_{-\lambda}, \qquad (gb, u) \sim (g, \chi_\lambda(b)^{-1}u).$$

Sections correspond to functions with

$$f(gb) = \chi_\lambda(b) f(g). \tag{26.4}$$

A section is holomorphic exactly when its function $f : G \to \mathbb{C}$ is holomorphic. Left translation acts as before. Write $H^0(G/B, L_\lambda)$ for the space of holomorphic sections of L_λ.

Theorem 26.4 (Borel–Weil). *With the preceding positive Borel and bundle convention, if λ is dominant integral, then*

$$H^0(G/B, L_\lambda) \cong V_\lambda^*$$

as a K-representation. This is finite-dimensional and irreducible. If an integral λ is not dominant, then $H^0(G/B, L_\lambda) = 0$.

The proof below uses the cited facts about B, χ_λ and G/B above, in particular $G = KB$. It also uses the holomorphic extension of Lecture 14 and the highest-weight theorem of the prerequisite course. The rest comes from the Peter–Weyl theorem.

The dual V_λ^* has highest weight $-w_0\lambda$. Here w_0 is the longest element of the Weyl group, the element that sends the positive roots to the negative roots. For $\mathrm{SU}(n)$, the element $-w_0$ reverses the Dynkin labels. For $\mathrm{SU}(2)$ we have $-w_0\lambda = \lambda$, so the dual is invisible there.

The case $\mathrm{SU}(3)$ by hand

Here $G = \mathrm{SL}(3,\mathbb{C})$, and B is the subgroup of upper triangular matrices. A *flag* is a line ℓ inside a plane P in $\mathbb{C}^3$. Figure 26.3 shows one. The group G moves flags, and B is the stabiliser of the standard flag $\mathbb{C}e_1 \subset \mathbb{C}e_1 \oplus \mathbb{C}e_2$. The group $K = \mathrm{SU}(3)$ already moves flags transitively, with stabiliser the diagonal torus T. So $K/T = G/B$ is the space of flags.

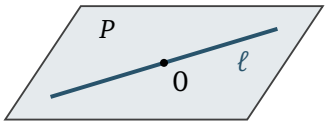


Figure 26.3: A flag $\ell \subset P$ in $\mathbb{C}^3$, drawn schematically: the complex line ℓ and the complex plane P appear as a real line in a real plane.

Example 26.5 (Sections for $\mathrm{SU}(3)$)**.** Let $\lambda = a\omega_1 + b\omega_2$ with integers $a, b \geq 0$, and write β for an upper triangular matrix in G. The character is trivial on the unipotent matrices. On the torus, ω_1 is the first diagonal entry and ω_2 is the product of the first two. So

$$\chi_\lambda(\beta) = \beta_{11}^{a+b}\beta_{22}^{b}.$$

Take a linear functional Φ on $\mathbb{C}^3$ and put $f(g) = \Phi(ge_1)$. Since $(g\beta)e_1 = \beta_{11}ge_1$, we get $f(g\beta) = \beta_{11}f(g)$. So f is a section for $(a,b) = (1,0)$. These sections form $(\mathbb{C}^3)^*$, the dual of $V_{(1,0)} = \mathbb{C}^3$, as the theorem says.

Next take a linear functional Ψ on $\Lambda^2\mathbb{C}^3$ and put $f(g) = \Psi(ge_1 \wedge ge_2)$. Since $(g\beta)e_1 \wedge (g\beta)e_2 = \beta_{11}\beta_{22}\, ge_1 \wedge ge_2$, this f is a section for $(0,1)$. These sections form the dual of $V_{(0,1)} = \Lambda^2\mathbb{C}^3$.

A product of a sections for $(1,0)$ and b sections for $(0,1)$ is a section for (a,b). These products need not be independent. For $(1,1)$, the nine products of coordinates of ge_1 with coordinates of $ge_1 \wedge ge_2$ satisfy one relation, since $ge_1 \wedge (ge_1 \wedge ge_2) = 0$. They span the eight-dimensional dual of the adjoint module.

Proof of the Borel–Weil theorem

Proof. We build sections from V_λ, and then we show that there are no others. Throughout, R_X denotes the derivative of a function φ on K along the right translation by $\exp tX$:

$$(R_X\varphi)(k) = \frac{d}{dt}\bigg|_0 \varphi(k \exp tX), \qquad X \in \mathfrak{k}.$$

We extend it complex-linearly to $Z \in \mathfrak{k}_{\mathbb{C}} = \mathfrak{g}$, with $R_{X+iY} = R_X + iR_Y$.

Step 1: sections from V_λ. Let λ be dominant. A module V_λ of highest weight λ exists by the highest-weight theorem of the prerequisite course and the integration theorem of Lecture 4. It extends to a holomorphic representation of G by Lecture 14. Let v_λ be a highest weight vector. Write Π_λ for this representation. On $\mathfrak{b}$, the operator $d\Pi_\lambda(Z)$ acts on v_λ by the scalar $d\chi_\lambda(Z)$: this is $i\lambda(H)$ for $H \in \mathfrak{t}$ and zero on the positive root spaces. So $b \mapsto \Pi_\lambda(b)v_\lambda$ and $b \mapsto \chi_\lambda(b)v_\lambda$ agree on $\exp \mathfrak{b}$, and hence on the connected group B:

$$\Pi_\lambda(b)v_\lambda = \chi_\lambda(b)v_\lambda.$$

For $\xi \in V_\lambda^*$ put $f_\xi(g) = \xi(\Pi_\lambda(g)v_\lambda)$. This function is holomorphic, and

$$f_\xi(gb) = \xi(\Pi_\lambda(g)\Pi_\lambda(b)v_\lambda) = \chi_\lambda(b)f_\xi(g).$$

So f_ξ is a holomorphic section of L_λ.

Step 2: the dual. Left translation by a gives

$$f_\xi(a^{-1}g) = \xi(\Pi_\lambda(a)^{-1}\Pi_\lambda(g)v_\lambda) = f_{\xi\circ\Pi_\lambda(a)^{-1}}(g).$$

So $\xi \mapsto f_\xi$ intertwines the dual representation with the section action. This is the reason for the dual: the section is linear in the functional ξ, and left translation acts on ξ. The map is injective. Indeed, if $f_\xi = 0$, then ξ vanishes on all $\Pi_\lambda(g)v_\lambda$. These vectors span V_λ because it is irreducible, so $\xi = 0$.

Step 3: the equations on K. Now let λ be any integral weight, and let f be holomorphic on G with the covariance (26.4). Let $\varphi = f|_K$, a smooth function on K. For $Z \in \mathfrak{g}$,

$$(R_Z\varphi)(k) = \left.\frac{d}{dt}\right|_0 f(k\exp tZ),$$

since the differential of the holomorphic function f is $\mathbb{C}$-linear. This is the point where holomorphy is used. A smooth function on K that is only covariant under T would not satisfy the second equation below. For $Z \in \mathfrak{b}$, the covariance gives $f(k\exp tZ) = e^{t\,d\chi_\lambda(Z)}f(k)$. Hence

$$R_H\varphi = i\lambda(H)\varphi \quad (H \in \mathfrak{t}), \qquad R_E\varphi = 0 \quad (E \text{ in a positive root space}). \tag{26.5}$$

Step 4: one block at a time. Let Π be an irreducible unitary representation of K, and let P_Π be the isotypic projector of Lecture 11. It is an integral of left translations:

$$(P_\Pi\varphi)(x) = d_\Pi \int_K \chi_\Pi(y)\,\varphi(y^{-1}x)\,dy.$$

Here χ_Π is the character of Π. The derivative $R_X\varphi$ is continuous on the compact group K, so we may differentiate under this integral, and $R_ZP_\Pi\varphi = P_\Pi R_Z\varphi$ for $Z \in \mathfrak{g}$. So $P_\Pi\varphi$ also satisfies (26.5).

The function $P_\Pi\varphi$ lies in the coefficient block M_Π. Let (ξ_i) be a basis of V_Π^*. Then

$$(P_\Pi\varphi)(k) = \sum_i \xi_i(\Pi(k)v_i), \qquad v_i \in V_\Pi,$$

and differentiating gives $(R_ZP_\Pi\varphi)(k) = \sum_i \xi_i(\Pi(k)d\Pi(Z)v_i)$. By Schur orthogonality (Lecture 5), the matrix coefficients are linearly independent: a sum $\sum_i \xi_i(\Pi(k)w_i)$ vanishes for all k only if every $w_i = 0$. So (26.5) gives

$$d\Pi(H)v_i = i\lambda(H)v_i, \qquad d\Pi(E)v_i = 0.$$

Thus each nonzero v_i is a highest weight vector of weight λ in the irreducible module V_Π. By the highest-weight theorem of the prerequisite course, if some $v_i \neq 0$, then λ is dominant and $\Pi \cong V_\lambda$. Its highest weight line is one-dimensional, so $v_i = c_i v_\lambda$, and

$$(P_\Pi \varphi)(k) = \xi(\Pi(k) v_\lambda), \qquad \xi = \sum_i c_i \xi_i .$$

For every other Π, we get $P_\Pi \varphi = 0$.

Step 5: completeness. We have not assumed that φ is a finite sum of coefficients. The Peter–Weyl theorem, in the form of the Plancherel theorem of Lecture 11, says that the sums of $P_\Pi \varphi$ converge to φ in $L^2(K)$. By Step 4 at most one term is nonzero. So φ equals that term in $L^2(K)$, and both functions are continuous, so they agree on K. If λ is not dominant, then $\varphi = 0$. If λ is dominant, then $\varphi = f_\xi|_K$ for some ξ.

Finally, $G = KB$, and the covariance determines a function on G from its restriction to K. So $f = 0$ when λ is not dominant, and $f = f_\xi$ when λ is dominant. With Step 2, this gives $H^0(G/B, L_\lambda) \cong V_\lambda^*$. The dual of an irreducible module is irreducible. ∎

The case $\mathrm{SU}(2)$

For $K = \mathrm{SU}(2)$ the group G is $\mathrm{SL}(2, \mathbb{C})$ (Lecture 14), and B is the subgroup of upper triangular matrices in G. The character is

$$\chi_m(b) = c^m, \qquad b = \begin{pmatrix} c & \beta \\ 0 & c^{-1} \end{pmatrix} \in B.$$

It is trivial on the unipotent matrices, where $c = 1$. On the diagonal matrices it is $\mathrm{diag}(c, c^{-1}) \mapsto c^m$, the character of weight m. The fibre at $\mathbb{C}(1,0)$ transforms by c^{-m}. The map $gB \mapsto [g(1,0)^T]$ identifies G/B with $\mathbb{CP}^1$. Indeed, G acts transitively on lines, since $\mathrm{SU}(2)$ already does, and B is the stabiliser of the line $\mathbb{C}(1,0)$.

We now compare the two section spaces directly. Let f be holomorphic on G with $f(gb) = \chi_m(b) f(g)$. The character is trivial on unipotent b, so f depends only on the first column $v = g(1,0)^T$. Put $F(v) = f(g)$. On $\{v_0 \neq 0\}$ and on $\{v_1 \neq 0\}$, the local sections

$$v \longmapsto \begin{pmatrix} v_0 & 0 \\ v_1 & 1/v_0 \end{pmatrix}, \qquad v \longmapsto \begin{pmatrix} v_0 & -1/v_1 \\ v_1 & 0 \end{pmatrix}$$

have determinant one and holomorphic entries. So F is holomorphic on $\mathbb{C}^2 \setminus \{0\}$. Also $F(cv) = c^m F(v)$ for $c \neq 0$, by the character on $\mathrm{diag}(c, c^{-1})$. Hence $f_0(z) = F(1, z)$ and $f_1(w) = F(w, 1)$ are entire, and they satisfy the transition rule (26.2). By Theorem 26.3, for $m \geq 0$ there is a homogeneous polynomial, again called F, of degree m with these two coefficients, so that $f = f_F$. For $m < 0$, the same theorem gives $f_0 = f_1 = 0$, so $f = 0$. Conversely, each $f_F(g) = F(g(1,0)^T)$ is holomorphic and covariant.

So $F \mapsto f_F$ is an isomorphism $H^0(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(G/B, L_m)$ of $\mathrm{SU}(2)$-representations for $m \geq 0$, and for $m < 0$ both spaces are zero. Hence our explicit calculation is exactly the case $\mathrm{SU}(2)$ of Borel–Weil. In Step 1 of the proof, $V_m = \mathrm{Sym}^m \mathbb{C}^2$ has highest weight vector $e_1^{\otimes m}$, with $e_1 = (1,0)^T$. So $f_\xi(g) = \xi((g e_1)^{\otimes m})$, which is the function f_F for the polynomial $F(v) = \xi(v^{\otimes m})$.

Self-checks

1. What is wrong with choosing an arbitrary entire f_0 as a section of $\mathcal{O}(1)$?

Answer. The coefficient $f_1(w) = w f_0(1/w)$ must also extend holomorphically to $w = 0$. This forces f_0 to have degree at most one.

2. Which representation occurs in $H^0(\mathbb{CP}^1, \mathcal{O}(3))$?

Answer. It is $\mathrm{Sym}^3(\mathbb{C}^2)^* \cong V_3$, of dimension four and spin $3/2$. The central element $-I$ acts by the scalar $(-1)^3 = -1$, so this representation does not descend to $\mathrm{SO}(3)$.

3. Does L_λ always produce highest weight λ in our general convention?

Answer. No. Its section space is V_λ^*, of highest weight $-w_0\lambda$, where w_0 is the longest element of the Weyl group. For $\mathrm{SU}(2)$ these weights coincide. For $\mathrm{SU}(n)$, the element $-w_0$ reverses the Dynkin labels, so for $\mathrm{SU}(3)$ the bundle $L_{(1,0)}$ gives highest weight $(0, 1)$.

Summary

Equivariant functions correspond to sections of associated bundles. On $\mathbb{CP}^1$ and for $m \geq 0$, the transition rule for $\mathcal{O}(m)$ identifies holomorphic sections with homogeneous polynomials of degree m. Compact induction contains $V_m, V_{m+2}, \ldots$, and holomorphy keeps only V_m. For compact, connected, simply connected, semisimple K, the Borel–Weil theorem gives $H^0(G/B, L_\lambda) \cong V_\lambda^*$ for dominant integral λ, and zero otherwise. We proved it from Peter–Weyl. Covariance and holomorphy make the vectors in each coefficient block highest weight vectors of weight λ, so only one block survives. The facts $G = KB$, the complex structure of G/B, the character χ_λ and the highest-weight theorem stay cited. For $\mathrm{SU}(3)$, first-column and wedge coordinates give explicit sections.

Sources and hypotheses

This section collects what the main line cites, with the checks of hypotheses and the matching of conventions. Nothing later depends on the order in which it is read.

Smooth structure of associated bundles. We show that $E = G \times_H V$ is a smooth complex vector bundle. The local sections s_i are smooth, so the maps $h_{ij} = s_i^{-1} s_j$ are smooth into G. They take values in the embedded subgroup H, so they are smooth into H by the codomain-restriction result for embedded submanifolds [C: Lee, Corollary 5.30]. The representation σ is smooth by the automatic-smoothness theorem of Lecture 3. Hence the transition functions $\sigma(h_{ij})$ are smooth. Let $k = \dim V$, choose a basis of V, and regard $\mathbb{C}^k$ as $\mathbb{R}^{2k}$. Each fibre of E is then a real vector space of dimension $2k$, and the real matrices of $\sigma(h_{ij})$ are smooth maps into $\mathrm{GL}(2k, \mathbb{R})$. The vector bundle chart lemma [C: Lee, Lemma 10.6] applies to these real data. It gives E a unique topology and smooth structure for which the inverses of the maps above are smooth local trivialisations. These maps are complex-linear on fibres, so E is a smooth complex vector bundle.

This structure does not depend on the chosen family of local sections. For a second family (s'_j), the maps $s_i^{-1}s'_j$ are smooth into H by the same codomain-restriction argument. So the mixed transition functions $\sigma(s_i^{-1}s'_j)$ are smooth, and the chart lemma applies to the union of the two families. By its uniqueness clause, the union gives the same structure as each family alone.

Regularity of covariant functions. If f is smooth, then each $f \circ s_i$ is smooth. Conversely, suppose that each $f \circ s_i$ is smooth. Write $\pi : G \to G/H$ for the projection, and put

$$q(g) = s_i(\pi(g))^{-1}g \qquad (g \in \pi^{-1}(U_i)).$$

This map is smooth into G and takes values in H, so it is smooth into the embedded subgroup H by the same codomain-restriction result. Since $g = s_i(\pi(g))q(g)$, the covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big).$$

Hence f is smooth on each open set $\pi^{-1}(U_i)$, and therefore on G. The same argument with continuity in place of smoothness shows that s is continuous exactly when f is continuous.

Holomorphic inverse functions. Let Φ be a map between open subsets of $\mathbb{C}^d$ with holomorphic components and invertible differential at a point. The real inverse function theorem gives a smooth local inverse. Its differential is the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. So the local inverse is holomorphic. This is the *holomorphic inverse function theorem*.

Holomorphic bundles. We work in the setting of the paragraph on holomorphic bundles in the main line. If $g = s_i(x)h$, then $R_h \circ s_i$ is a holomorphic, hence smooth, local section through g, because $\pi \circ R_h = \pi$. By the local-section theorem of Lee, Theorem 4.26, π is a smooth submersion.

The topology of G/H is the quotient topology. Indeed, π is continuous, since it is holomorphic. Conversely, let $U \subset G/H$ be a set such that $\pi^{-1}(U)$ is open. Then $U \cap U_i = s_i^{-1}(\pi^{-1}(U))$ is open for each i, because $\pi \circ s_i = \mathrm{id}$. The sets U_i cover G/H, so U is open. By the uniqueness of the quotient structure in Lecture 3, the smooth structure underlying the complex manifold G/H is the quotient smooth structure.

Over the domain of a holomorphic chart φ of G/H inside U_i, the map $[s_i(x), v] \mapsto (\varphi(x), v)$ is a chart of E, with V identified with $\mathbb{C}^k$ by a basis. The change between two such charts has the form $(y, u) \mapsto (\psi(y), A(y)u)$. Here ψ is a transition map of G/H, and $A(y)$ is the matrix of $\sigma(h_{ij})$ at the point with coordinate y. The entries of A are holomorphic, so this change is holomorphic. Hence these charts make E a complex manifold. For a second family (s'_j) of holomorphic local sections, the mixed changes of chart have the same form with $\sigma(s_i^{-1}s'_j)$, which is holomorphic by the slice-chart property. Hence both families give one holomorphic atlas, and the complex structure of E does not depend on the family.

Complex analysis. [C: Stein and Shakarchi, *Complex Analysis*, 2003, Chapter 2, Section 4]. Theorem 4.4 and the paragraph after it state that an entire function has a power series about 0 that converges on all of $\mathbb{C}$, with $a_k = f^{(k)}(0)/k!$. The Cauchy inequalities, Corollary 4.3, give

$$|f^{(k)}(0)| \le k! R^{-k} \max_{|z|=R} |f(z)|.$$

Dividing by $k!$ gives the estimate for a_k.

The matrix model of $K_{\mathbb{C}}$. Sepanski, *Compact Lie Groups*, states Borel–Weil for a matrix model of G. We identify the two groups. A faithful representation ρ of K exists (Lecture 10; also [C: Bröcker and tom Dieck (BD), Section III.4, Theorem 4.1]). Averaging makes ρ unitary (Lecture 5). The entries of $r = \rho \oplus \overline{\rho} : K \to \mathrm{U}(N)$ generate $\mathscr{R}(K)$ [C: BD, Section III.4, Proposition 4.3].

Recall from Lecture 14 that each $\psi \in K_{\mathbb{C}} = \mathcal{X}(K)$ is a unital algebra homomorphism from $\mathscr{R}(K)$ to $\mathbb{C}$. The *coefficient map* $r_{\mathbb{C}}$ evaluates ψ on the matrix entries r_{ij} of r:

$$r_{\mathbb{C}}(\psi) = \big(\psi(r_{ij})\big)_{i,j=1}^{N}.$$

By [C: BD, Section III.8, Proposition 8.2 and the next paragraph], the map $r_{\mathbb{C}}$ sends $K_{\mathbb{C}}$ biholomorphically onto a closed subgroup and analytic submanifold of $\mathrm{GL}(N,\mathbb{C})$.

The resulting complex structure on $K_{\mathbb{C}}$ does not depend on r. Let r' be a second faithful unitary representation of K whose entries also generate $\mathscr{R}(K)$. Let $r'_{\mathbb{C}}$ be its coefficient map, defined by the same construction. Each entry of r' is a polynomial in the entries of r, and conversely. Every $\psi \in K_{\mathbb{C}}$ is a unital algebra homomorphism, so it respects these polynomial relations. Hence $r'_{\mathbb{C}} \circ r_{\mathbb{C}}^{-1}$ and its inverse are restrictions of polynomial maps. By the slice-chart property, they are holomorphic between the two analytic submanifolds. This common structure is the complex structure of Lecture 14.

The Lie algebra $\mathfrak{g}$ of $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complex span of that of $r(K)$ [C: BD, Proposition 8.4 and the preceding paragraph].

Since K is connected, so is $K_{\mathbb{C}}$ (Lecture 14). A connected Lie subgroup of $\mathrm{GL}(N,\mathbb{C})$ is determined by its Lie algebra [C: Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 3.40]. Hence $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complexification of $r(K)$ in Sepanski, Definition 7.48. We identify G with $r_{\mathbb{C}}(K_{\mathbb{C}})$ and K with $r(K)$, since $r_{\mathbb{C}}(\mathrm{ev}_g) = r(g)$.

Sepanski's construction applies to every faithful unitary r.

The Borel subgroup and the quotient. By Sepanski, Definition 7.48, B is the connected Lie subgroup of G with Lie algebra $\mathfrak{b}$. By Sepanski, Lemma 7.49, B is closed. By Sepanski, Theorem 7.50, the inclusion induces a diffeomorphism from K/T onto G/B.

Since B is closed, it is an embedded Lie subgroup of G by the closed-subgroup theorem of Lecture 3. The inclusion of B with Sepanski's Lie subgroup structure is smooth into this embedded submanifold by Lee, Corollary 5.30. It is a bijective Lie group homomorphism, so it is a Lie group isomorphism by Lee, Corollary 7.6. Hence the embedded group B has Lie algebra $\mathfrak{b}$, and exp maps some neighbourhood O of 0 in $\mathfrak{b}$ diffeomorphically onto an open subset of B.

Choose a complex complement $\mathfrak{m}$ of $\mathfrak{b}$ in $\mathfrak{g}$. For $b \in B$, define $\Phi : \mathfrak{b} \times \mathfrak{m} \to G$ by

$$\Phi(X, Y) = b \exp X \exp Y.$$

The matrix exponential is holomorphic and maps $\mathfrak{g}$ into G, so Φ is holomorphic into G by the slice-chart property. Its differential at $(0, 0)$ is the complex-linear isomorphism $(X, Y) \mapsto b(X + Y)$. By the holomorphic inverse function theorem, after shrinking O, there is a neighbourhood O' of 0 in $\mathfrak{m}$ such that Φ maps $O \times O'$ biholomorphically onto an open set $W' \subset G$. The set $\Phi(O \times \{0\}) = b \exp O$ is open in B. Because B has the subspace topology, it equals $B \cap W''$ for some open set W''. On $W = W' \cap W''$, the set B is exactly the slice $Y = 0$. Hence B is a closed complex subgroup.

The complex quotient theorem [C: Kirillov Jr., Theorem 2.11 and its proof] makes G/B a complex manifold, with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections. Thus the first section applies to G, B and G/B.

Weights and the source of the theorem. Our character of weight λ is $\exp H \mapsto e^{i\lambda(H)}$, so Sepanski's weight is our $i\lambda$ on $\mathfrak{t}$ (Sepanski, Definitions 6.23 and 6.26, Theorem 6.27 and formula (6.28)). In our labels, Sepanski, Definition 7.52, defines the character χ_λ of the main line. Sepanski's Theorem 7.58 states the Borel–Weil theorem for the groups G and B above. His Definition 7.54 calls a section holomorphic when its function on G is holomorphic, and by the equivalence in the main line his holomorphic sections are ours. Our L_λ is his line bundle of index $-\lambda$, so his theorem gives the same V_λ^*. Our proof does not use his theorem.

The group $\mathrm{SL}(2, \mathbb{C})$ as $\mathrm{SU}(2)_{\mathbb{C}}$. For $\mathrm{SU}(2)$, take r to be the standard representation. Its four entries generate the representative functions, and $r_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{SL}(2, \mathbb{C})$ (Lecture 14). We give $\mathrm{SL}(2, \mathbb{C})$ its complex structure as the subset $\{\det = 1\}$ of $\mathbb{C}^4$. This subset is a complex submanifold, with explicit slice charts. On the open set $\{g_{11} \neq 0\} \subset \mathbb{C}^4$, put

$$\varphi(g_{11}, g_{12}, g_{21}, g_{22}) = (g_{11}, g_{12}, g_{21},\ g_{11}g_{22} - g_{12}g_{21} - 1).$$

The map φ is holomorphic onto the open set $\{(a, b, c, t) : a \neq 0\}$, with holomorphic inverse

$$(a, b, c, t) \longmapsto \big(a, b, c, (t + 1 + bc)/a\big).$$

In this chart, $\mathrm{SL}(2, \mathbb{C})$ is the set $t = 0$. On $\{g_{12} \neq 0\}$, $\{g_{21} \neq 0\}$ and $\{g_{22} \neq 0\}$, solve $\det g = 1 + t$ in the same way for the entry that multiplies the nonzero one in $\det g$. These four open sets cover $\mathrm{SL}(2, \mathbb{C})$, because $\det g = 1$ forces some entry of g to be nonzero.

By the slice-chart property, a map into $\mathrm{SL}(2, \mathbb{C})$ with holomorphic entries is holomorphic. On $\mathrm{SL}(2, \mathbb{C})$ the inverse is

$$g^{-1} = \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}.$$

Hence multiplication and inversion have polynomial entries, and $\mathrm{SL}(2, \mathbb{C})$ is a complex Lie group.

We now compare this structure with the complex structure of $\mathrm{SU}(2)_{\mathbb{C}} = \mathcal{X}(\mathrm{SU}(2))$ from Lecture 14. The example for $\mathrm{SU}(2)$ in Lecture 14 shows that the four matrix entries D_{ij} of

the standard representation generate the representative functions of SU(2). It also gives the bijection

$$P : \mathrm{SU}(2)_{\mathbb{C}} \longrightarrow \mathrm{SL}(2,\mathbb{C}), \qquad P(\psi) = \big(\psi(D_{ij})\big)_{i,j=1,2}.$$

The formula $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$ gives $P(\psi_1\psi_2) = P(\psi_1)P(\psi_2)$, so P is a group isomorphism. Also $P(\mathrm{ev}_g) = g$ for $g \in \mathrm{SU}(2)$. In the notation of Bröcker and tom Dieck, P is the map $r_{\mathbb{C}}$ for the standard representation r of SU(2).

Cited result. We apply [C: Bröcker and tom Dieck, *Representations of Compact Lie Groups*, Section III.8, Proposition 8.2 and the paragraph after it] to the standard representation r, whose entries generate the representative functions. Proposition 8.2 states that $r_{\mathbb{C}} = P$ maps $\mathrm{SU}(2)_{\mathbb{C}}$ bijectively onto an affine variety in $\mathbb{C}^4$. This variety is the image $\mathrm{SL}(2,\mathbb{C})$ of P. The paragraph after it states that the topology of $\mathrm{SU}(2)_{\mathbb{C}}$ is the subspace topology induced by P. It also states that the variety is an analytic submanifold of $\mathbb{C}^4$, on which multiplication and inversion are holomorphic. With this structure, P is a biholomorphic group isomorphism from $\mathrm{SU}(2)_{\mathbb{C}}$ onto $\mathrm{SL}(2,\mathbb{C})$ that is the identity on SU(2).

This analytic-submanifold structure on $\mathrm{SL}(2,\mathbb{C})$ coincides with the slice-chart structure above. Indeed, the inclusion of $\mathrm{SL}(2,\mathbb{C})$ into $\mathbb{C}^4$ is holomorphic for each structure. By the slice-chart property of the other structure, the identity map of $\mathrm{SL}(2,\mathbb{C})$ is holomorphic in each direction.

Here we use the standard representation r. We identify $\mathrm{SU}(2)_{\mathbb{C}}$ with $G = \mathrm{SL}(2,\mathbb{C})$ through P, which is the identity on SU(2). By the polynomial argument above, the complex structure of Lecture 14 does not depend on the generating representation. So it is the structure transported by P, which is $r_{\mathbb{C}}$ for the standard representation. By the cited result and the preceding paragraph, this structure coincides with the submanifold structure on G.

Consider the maps

$$s_0(z) = \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix}, \qquad s_1(w) = \begin{pmatrix} w & -1 \\ 1 & 0 \end{pmatrix}.$$

Both matrices have determinant one. Their first columns are the tautological frames $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. Identify G/B with $\mathbb{CP}^1$ by $gB \mapsto [g(1,0)^T]$, as in the main line. Then s_0 and s_1 are local sections of $G \to G/B$ over U_0 and U_1. Their entries are polynomials, so they are holomorphic maps into G.

At a point of the overlap, $w = 1/z$ and

$$h_{01} = s_0(z)^{-1} s_1(1/z) = \begin{pmatrix} 1 & 0 \\ -z & 1 \end{pmatrix} \begin{pmatrix} z^{-1} & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} z^{-1} & -1 \\ 0 & z \end{pmatrix} \in B.$$

Hence $\chi_m(h_{01}) = z^{-m}$.

Let $f : G \to \mathbb{C}$ satisfy $f(gb) = \chi_m(b) f(g)$, and put $f_0 = f \circ s_0$ and $f_1 = f \circ s_1$. Since $s_1(1/z) = s_0(z) h_{01}$, covariance gives

$$f_1(1/z) = f\big(s_0(z) h_{01}\big) = z^{-m} f_0(z) \qquad (z \neq 0).$$

With $w = 1/z$, this is the transition rule $f_1(w) = w^m f_0(1/w)$ of $\mathcal{O}(m)$.

Next we compare the two complex structures on G/B. Write $\pi : G \to G/B$ for the projection, and identify G/B with $\mathbb{CP}^1$ by $gB \mapsto [g(1,0)^T]$. The vector $g(1,0)^T$ is the first column

$(g_{11}, g_{21})^T$ of g. Hence $\pi^{-1}(U_0) = \{g_{11} \neq 0\}$ and $\pi^{-1}(U_1) = \{g_{21} \neq 0\}$ are open in G. In the charts,

$$z \circ \pi(g) = \frac{g_{21}}{g_{11}} \quad \text{on } \pi^{-1}(U_0), \qquad w \circ \pi(g) = \frac{g_{11}}{g_{21}} \quad \text{on } \pi^{-1}(U_1).$$

These are quotients of holomorphic functions with nonvanishing denominators. Hence π is holomorphic for the chart structure. The sections s_0 and s_1 are holomorphic for the chart structure, because their entries are polynomials. By the comparison of the two group structures above, the cited structure on G/B also makes π holomorphic for the submanifold structure on G, and it has holomorphic local sections.

Let s be a holomorphic local section for one of the two structures. Then $\pi \circ s$ is the identity map on the domain of s. This composition is holomorphic into the other structure, because π is holomorphic for that structure. Applying this in both directions shows that the identity map of G/B is holomorphic with holomorphic inverse. Hence the two complex structures coincide.

Let s' be another holomorphic local section over an open subset of U_0. Then $q = s_0^{-1}s'$ takes values in B, and it is holomorphic into B by the slice-chart property. Covariance gives $f \circ s' = \chi_m(q) f_0$. The factor $\chi_m(q)$ is holomorphic and nowhere zero, so it does not change holomorphy. The same holds over U_1. Hence a section of L_m is holomorphic exactly when f_0 and f_1 are entire.

This is the agreement of the two notions of holomorphic section that the projective-line calculation uses.

Reading

- M. R. Sepanski, *Compact Lie Groups* (2007), Section 7.4: the Borel–Weil theorem and the groups G and B.
- J. M. Lee, *Introduction to Smooth Manifolds*, second edition (2013), Chapters 4, 5 and 10: local sections, embedded submanifolds and the vector bundle chart lemma.
- T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (1985), Sections III.4 and III.8: faithful representations and complexification.